While recent methods in NLP have led to many significant and exciting empirical breakthroughs, they have often done so with increasingly expensive training data sizes and inference times. Perhaps as a pessimistic indicator of this research direction's upper bound in performance, such methods still have difficulty demonstrating key human-like abilities such as capturing the compositional aspect of language. Efficiency and compositionality are interrelated as current approaches often use over-parameterization and large training sets to hide their inability to exploit higher-level compositional properties. To get around these cost constraints and expand the capabilities of NLP, in my PhD at Stanford, I plan to design new architectures and methods that tackle these core limitations regarding compositionality, while ensuring they remain both parameter and data-efficient. Specifically, I am interested in the following two directions:

1. **Efficient NLP**: Given real-world resource constraints, how can we develop models that are more data-efficient as well as parameter-efficient for training and deployment?

2. **Compositional Generalization**: How can we make models generalize better with limited training examples? Can we draw inspiration from structural generalization, such as compositional structures of human language and cognition? [1]

Preparedness. My first and enjoyable experience with computer science was participating in high school programming olympiads, for which I was two-time national champion for USACO Open, and selected as 1 of the 4 team members to represent USA in International Olympiads in Informatics (IOI), the only female over the last 13 years. Throughout college, I have been fortunate to pursue multiple research experiences which have prepared me for PhD studies. As a freshman, I worked on the DataCalculator project with Prof. Stratos Idreos, which enables interactive and semi-automated design of data structures. This work appeared at SIGMOD 2018, and I was awarded Honorable Mention (top 20) for NCWIT 2018. In sophomore year, I worked with Prof. Samuel Gershman and Dr.Dasgupta to investigate the differences between human and computer generalization behaviors, focusing on the compositionality of sentence embeddings. The work was published at NAACL 2018 Gen-Deep workshop, CogSci 2018, and recently *Cognitive Science* (in press). I took a gap year to work at Facebook AI Research (FAIR) as a Research Engineer, where I joined three team efforts: Minecraft assistant bot, BERT for protein sequences, and 3D self-supervised learning. These projects led to a co-first-authored ICCV paper, a co-first-authored BioXiv manuscript, and an ECCV publication. [2]

Both of my research interests were developed in these works. My experience in systems building with Prof. Idreos and in working with projects utilizing significant computational resources at FAIR, have inspired an interest in efficient modeling. My work with Prof. Gershman instilled great curiosity about refining the human-like capabilities of natural language models.

I therefore became increasingly passionate about enhancing competence and resource-efficiency of NLP models, and started conducting preliminary work with Prof. Alexander Rush and Prof. Yoon Kim in these two directions:

Efficient NLP. Recently, a combination of scale in data and model capacity has led to advances in NLP. However, training such models can be impractical and unsustainable as it is cost and resource intensive. In many important tasks such as device-based NLP, resources are limited, further restricting the adoption of these advances. In particular, while task-specific finetuning of deep networks is necessary, their large size makes the standard finetuning approach difficult to deploy to multi-task, memory-constrained settings, as storing the full model parameters for each task becomes prohibitively expensive.

Motivated by this challenge, I propose an approach for more parameter-efficient finetuning (under review at ICLR 2021) [1]. Our approach, dubbed *diff pruning*, views task-specific finetuning as learning a diff vector $\boldsymbol{\delta}_\tau$ on top of the pretrained model parameters $\boldsymbol{\theta}_{\text{pretrained}}$ which remain fixed and are shared across different tasks. The diff vector is additionally regularized with a differentiable approximation to the L_0-norm penalty to encourage sparsity. Formally, for each task τ we reparameterize the task-specific model parameters as $\boldsymbol{\theta}_\tau = \boldsymbol{\theta}_{\text{pretrained}} + \boldsymbol{\delta}_\tau$, and the new training objective can be written as:

$$\min_{\boldsymbol{\delta}_\tau} \frac{1}{N} \sum_{n=1}^{N} \mathcal{L}\big(f(x_\tau^{(n)}; \boldsymbol{\theta}_{\text{pretrained}} + \boldsymbol{\delta}_\tau), y_\tau^{(n)}\big) + \lambda \|\boldsymbol{\delta}_\tau\|_0$$

While it is hard to optimize the L_0-norm penalty above directly, we found that using a differentiable approximation [2] was empirically effective. As the number of tasks increases, this approach can become particularly parameter-efficient, as it only requires nonzero parameters of the diff vector for each task. On both the GLUE and SQuAD benchmark, *diff pruning* can match the performance of the fully finetuned BERT baselines while finetuning only 0.5% of the pretrained

[1] Consider a classic example of a model which fails to exhibit compositionality: the model can recognize individual cats and tables, recognize the color black, and count objects—but is ironically unable to perform a task such as counting how many black cats are under the table.

[2] Please refer to my CV for publications details.

parameters per task. It outperforms the previous best alternative *adapters* [3] which requires 2.0% new parameters to achieve the same accuracy.

Beyond memory-efficiency, I am also interested in faster pre-training models [4] and inference methods [5]. I hope to additionally explore ways to integrate data systems and computer architecture designs into model training and inference, for which my experience in systems research as well as engineering can be potentially beneficial. Through pursuing this line of research, I am excited to make technology more efficient, increase practical deployment of models, and contribute to the democratization of research.

Compositional Generalization. Compositional generalization is also crucial to efficient NLP since compositional models are often much more data-efficient. Compositionality is one reason why humans can efficiently learn from a few examples and easily adapt to novel settings. Neural models, however, often have difficulty capturing such principles [6].

One approach to improving compositionality is through data augmentation [7,8]. Inspired by previous work, I worked on a data augmentation method to offer compositional inductive bias for sequence to sequence problems (EMNLP 2020) [9]. We specifically considered systematic compositionality of language, which implies that a computational model should understand sentences by appropriately recombining subparts that have not been seen together during training. For example, if a model understands "brown dogs" and "cute cats", it should also understand "cute dogs". Therefore, an intuitive solution to encourage compositionality is to create synthetic data examples that randomly recombine parts of the training sentences. Given the infinite possibilities of recombinations, we instead propose a soft variant of this augmentation strategy named *SeqMix*. Formally, we model the hard recombination process for two training examples (X, Y) and (X', Y') with a binary mask vector m_X, m_Y sampled from a random distribution. A new example can then be written as $(\hat{X}, \hat{Y}) = (m_X \odot X + (1 - m_X) \odot X', m_Y \odot Y + (1 - m_Y) \odot Y')$. Instead of enumerating over (or sampling) different masks, the soft variant uses the expectation of masks to create additional examples. Let $f_\theta(X, Y_{<t})$ be the output of the log-softmax layer, and the local probability of Y_t is formulated as $\log p_\theta(Y_t|X, Y_{<t}) = Y_t^\top f_\theta(X, Y_{<t})$. The final training objective becomes:

$$\mathcal{L} \approx \mathop{\mathbb{E}}_{\substack{(X,Y)\sim D \\ (X',Y')\sim D'}} \left[\sum_{t=1}^{T} \mathbb{E}[\hat{Y}_t]^\top f_\theta \left(\mathbb{E}[\hat{X}], \mathbb{E}[\hat{Y}_{<t}] \right) \right]$$

While *SeqMix* is a crude approximation to language-like compositional phenomena, empirically it can improve across different tasks (i.e. machine translation, semantic parsing and SCAN [6]), and exhibits non-trivial compositionality.

Apart from data augmentation, I am interested in other methods to enhance compositionality, such as modularizing networks [10], and integrating neural-symbolic systems [11]. Furthermore, I hope to conduct research to better measure and advance compositionality in more naturalistic and general setups: How could we formulate, evaluate and improve compositionality both in diagnostic datasets and in complex real-world applications? Can we develop compositional methods that are more widely applicable to different tasks? I am excited to tackle these questions, drawing insights not only from NLP but also from biology, cognitive science and philosophy.

Future Plan. Past experiences have led to my interests in efficiency and generalization, yet I am also open to other important problems in NLP and machine learning. I am eager to attend Stanford's PhD program, because of the many professors whose work greatly interests me, including Percy Liang, Christopher Manning and Tatsunori Hashimoto. As an astute consumer of the research publications produced by these faculty members, I see a clear fit for my skills and interests at Stanford and I am confident that I will thrive and contribute to the Computer Science Department.

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